

Smart Gym & Fitness Centre Management System

1. Data Dictionary

Table 1: users

Purpose: Core user authentication and profile management for all system users

Column	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY	Unique identifier for each user
email	STRING	NOT NULL, UNIQUE	Login email address
password_hash	STRING	NOT NULL	Encrypted password
first_name	STRING	NOT NULL	User's first name
last_name	STRING	NOT NULL	User's last name
phone	STRING		Contact phone number
role	STRING	NOT NULL	User role in the system
date_of_birth	STRING		Birth date
gender	STRING		Gender identity
created_at	STRING		Account creation time
is_active	BOOL		Account active status

Table 2: membership_tiers

Purpose: Defines membership tier offerings (Basic, Premium, VIP)

Column	Data Type	Constraints	Description
tier_id	INT	PRIMARY KEY	Unique tier identifier
tier_name	STRING	NOT NULL, UNIQUE	Membership tier name
monthly_fee	FLOAT	NOT NULL	Monthly subscription cost

access_sauna	BOOL		Sauna access permission
access_pool	BOOL		Pool access permission
access_premium_zone	BOOL		Premium area access
max_classes_per_month	INT		Allowed class bookings
description	STRING		Tier benefits description

Table 3: memberships

Purpose: Tracks active member subscriptions and membership lifecycle

Column	Data Type	Constraints	Description
membership_id	INT	PRIMARY KEY	Unique membership identifier
user_id	INT	FOREIGN KEY	Member reference
tier_id	INT	FOREIGN KEY	Membership tier reference
start_date	STRING	NOT NULL	Membership start date
end_date	STRING	NOT NULL	Membership expiry date
is_frozen	BOOL		Freeze status
frozen_days_remaining	INT		Remaining freeze days
auto_renew	BOOL		Auto-renewal flag
status	STRING		Membership status

Additional Constraint: CHECK(end_date > start_date) — Ensures valid date range

Table 4: trainers

Purpose: Trainer profiles with specialisations (1:1 relationship with users)

Column	Data Type	Constraints	Description
trainer_id	SERIAL	PRIMARY KEY	Unique trainer identifier

user_id	INTEGER	FOREIGN KEY → users(user_id), UNIQUE, NOT NULL	Links to users table (1:1)
specialization	VARCHAR(100)		Training specialization areas
certification	VARCHAR(255)		Professional certifications
hourly_rate	DECIMAL(10,2)	CHECK(hourly_rate >= 0)	Session rate (PKR)
years_experience	INTEGER	CHECK(years_experience >= 0)	Years of experience
bio	TEXT		Trainer biography
is_available	BOOLEAN	DEFAULT TRUE	Availability status
rating	DECIMAL(3,2)	CHECK(rating >= 0 AND rating <= 5)	Average customer rating

Table 5: classes

Purpose: Group class definitions/templates

Column	Data Type	Constraints	Description
class_id	INT	PRIMARY KEY	Unique class identifier
class_name	STRING	NOT NULL	Class name
description	STRING		Class details
class_type	STRING		Category type
duration_minutes	INT	NOT NULL	Class duration
difficulty_level	STRING		Difficulty level
max_capacity	INT	NOT NULL	Maximum participants

Table 6: class_schedules

Purpose: Scheduled instances of classes with a specific date/time/trainer

Column	Data Type	Constraints	Description
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schedule_id	INT	PRIMARY KEY	Unique schedule identifier
class_id	INT	FOREIGN KEY	Class reference
trainer_id	INT	FOREIGN KEY	Assigned trainer
schedule_date	STRING	NOT NULL	Class date
start_time	STRING	NOT NULL	Start time
end_time	STRING	NOT NULL	End time
room_number	STRING		Room location
current_capacity	INT		Current bookings
status	STRING		Schedule status

Additional Constraint: CHECK(end_time > start_time) — Ensures valid time range

Table 7: class_bookings

Purpose: Junction table for many-to-many relationship between users and class schedules

Column	Data Type	Constraints	Description
booking_id	INT	PRIMARY KEY	Unique booking identifier
schedule_id	INT	FOREIGN KEY	Scheduled class reference
user_id	INT	FOREIGN KEY	Member reference
booking_date	STRING		Booking time
status	STRING		Booking status
position_in_waitlist	INT		Waitlist position

Additional Constraint: UNIQUE(schedule_id, user_id) — Prevents duplicate bookings

Table 8: workout_plans

Purpose: Trainer-created workout templates assigned to members

Column	Data Type	Constraints	Description
plan_id	INT	PRIMARY KEY	Unique plan identifier
user_id	INT	FOREIGN KEY	Assigned member
trainer_id	INT	FOREIGN KEY	Plan creator
plan_name	STRING	NOT NULL	Plan name
description	STRING		Plan details
goal	STRING		Fitness goal
start_date	STRING	NOT NULL	Plan start date
end_date	STRING		Plan end date
is_active	BOOL		Active status
created_at	STRING		Creation timestamp

Table 9: workout_logs

Purpose: Individual exercise logging by members

Column	Data Type	Constraints	Description
log_id	INT	PRIMARY KEY	Unique log identifier
user_id	INT	FOREIGN KEY	Member logging workout
plan_id	INT	FOREIGN KEY	Related plan
workout_date	STRING	NOT NULL	Workout date
exercise_name	STRING	NOT NULL	Exercise name
muscle_group	STRING		Target muscle
sets	INT		Number of sets
reps	INT		Repetitions
weight_kg	FLOAT		Weight used

duration_minutes	INT		Exercise duration
calories_burned	INT		Calories burned
difficulty	STRING		Difficulty level
notes	STRING		Personal notes
created_at	STRING		Log timestamp

Table 10: equipment

Purpose: Gym equipment inventory and status tracking

Column	Data Type	Constraints	Description
equipment_id	INT	PRIMARY KEY	Equipment identifier
equipment_name	STRING	NOT NULL	Equipment name
equipment_type	STRING		Equipment category
zone	STRING		Gym area
purchase_date	STRING		Purchase date
purchase_cost	FLOAT		Cost
manufacturer	STRING		Brand
model_number	STRING		Model info
status	STRING		Equipment status
total_usage_hours	INT		Usage hours
last_maintenance_date	STRING		Last service date
next_maintenance_due	STRING		Next service date

Table 11: equipment_maintenance

Purpose: Equipment maintenance and repair history

Column	Data Type	Constraints	Description
maintenance_id	INT	PRIMARY KEY	Maintenance identifier
equipment_id	INT	FOREIGN KEY	Equipment reference
maintenance_date	STRING	NOT NULL	Service date
maintenance_type	STRING		Maintenance type
description	STRING		Work details
technician_name	STRING		Technician
cost	FLOAT		Maintenance cost
status	STRING		Maintenance status
created_at	STRING		Record creation time

Table 12: lockers

Purpose: Locker inventory and assignment tracking

Column	Data Type	Constraints	Description
locker_id	INT	PRIMARY KEY	Unique locker identifier
locker_number	STRING	UNIQUE	Locker number
locker_size	STRING		Size of locker (small/medium/large)
location	STRING		Physical locker location
status	STRING		Current locker status
current_user_id	INT	FOREIGN KEY	User currently assigned

assigned_at	STRING		Assignment timestamp
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Table 13: attendance

Purpose: Member check-in/check-out tracking for access control

Column	Data Type	Constraints	Description
attendance_id	INT	PRIMARY KEY	Unique attendance record
user_id	INT	FOREIGN KEY	Member checking in
check_in_time	STRING	NOT NULL	Entry time
check_out_time	STRING		Exit time
duration_minutes	INT		Time spent in gym
locker_assigned	INT	FOREIGN KEY	Locker used during visit
access_areas	STRING		Areas accessed inside gym

Additional Constraint: CHECK(check_out_time IS NULL OR check_out_time > check_in_time)

Table 14: payments

Purpose: All financial transactions (memberships, sessions, products, fees)

Column	Data Type	Constraints	Description
payment_id	INT	PRIMARY KEY	Unique payment identifier
user_id	INT	FOREIGN KEY	User making payment
payment_type	STRING		Type of payment
amount	FLOAT	NOT NULL	Payment amount
payment_method	STRING		Method used for payment
payment_date	STRING		Transaction date

reference_id	INT		Related record reference
description	STRING		Payment details
status	STRING		Payment status

Table 15: nutrition_plans

Purpose: Trainer-created meal plans for members

Column	Data Type	Constraints	Description
nutrition_plan_id	INT	PRIMARY KEY	Unique nutrition plan identifier
user_id	INT	FOREIGN KEY	Member assigned to plan
trainer_id	INT	FOREIGN KEY	Trainer who created plan
plan_name	STRING	NOT NULL	Name of nutrition plan
daily_calories_target	INT		Daily calorie goal
protein_grams	INT		Protein intake target
carbs_grams	INT		Carbohydrate intake target
fats_grams	INT		Fat intake target
meals_plan	STRING		Meal details
start_date	STRING	NOT NULL	Plan start date
end_date	STRING		Plan end date
is_active	BOOL		Active plan status
created_at	STRING		Creation timestamp

2. INDEXES

Index Name	Type	Table	Columns	Purpose
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idx_payments_user_date	Composite	payments	(user_id, payment_date)	Billing reports & payment history
idx_attendance_checkin	Single-column	attendance	(check_in_time)	Attendance tracking & time reports
idx_schedules_date_time	Composite	class_schedules	(schedule_date, start_time)	Class availability search

3. USER ROLES

Role	Access Level	Key Permissions
member	Limited (own data only)	Read own profile/membership/bookings; book & log workouts
trainer	Assigned clients & classes	Create workout/nutrition plans; manage class schedules
staff	Operational daily gym ops	Check-in/out members; assign lockers; update equipment status
admin	Full system access	User management, tier config, financial reports, all data

5. NORMALIZATION

Normal Form	Key Points
1NF	Atomic values; no repeating groups (JSON exception in access_areas)
2NF	All single-column surrogate PKs; no partial dependencies
3NF	No transitive dependencies; proper table separation

Justified Denormalization (3 cases):

- current_capacity in class_schedules (performance, updated via trigger)

- total_usage_hours in equipment (maintenance scheduling, updated via trigger)
- duration_minutes in attendance (reporting performance, calculated at check-out)

APPENDIX: RELATIONSHIPS

1:1 — Users ↔ Trainers (via trainers.user_id UNIQUE)

1:M — Users→Memberships, Users→WorkoutLogs, Users→Attendance, Users→Payments, MembershipTiers→Memberships, Trainers→WorkoutPlans, Trainers→NutritionPlans, Trainers→ClassSchedules, Classes→ClassSchedules, ClassSchedules→ClassBookings, WorkoutPlans→WorkoutLogs, Equipment→EquipmentMaintenance, Lockers→Attendance

M:M — Users ↔ ClassSchedules (via class_bookings junction table)